CT PERFORMANCE EVALUATION REPORT

PRE-TEST / POST FAN CHANGE / POST DISTRIBUTION CHANGE

— THREE-STAGE SINGLE CELL ASSESSMENT —

CT - 02 , CELL 02

OWNER / CLIENT	DHARIWAL INFRASTRUCTURE LTD CHANDRAPUR
TESTING AGENCY	SS COOLING TOWER CONSULTANTS (SSCTC)
DATE OF TEST	27 March 2026
REPORT DATE	15 April 2026
ASSESSMENT METHOD	CTI ATC-105 Merkel Method — Applied independently to each of 3 test stages

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PREAMBLE

Due to poor performance of the cooling tower, factors influencing the poor performance were identified to be progressively resolved. To carry out this procedure it was decided to carry out the changes on one cell and based on the result carry out the same on balance cell. This procedure had to be conducted both before change of gear box and fan and post change of gear box and fan to ascertain the effect on cold water temperature.

MEMBERS PRESENT FOR THE TEST

CLIENT

- Mr Shrikant Shrivastava (Remote)
- Mr GAURAV HANTODKAR
- Mr PAWAN GAWANDE

SS COOLING TOWER CONSULTANTS (SSCTC)

- Mr SURESH SARMA
- Mr SANJAY GORAD
- Mr MRADUL VISHWAKARMA
- Mr PARAG VISHWAKARMA
- Mr RAHUL MANKE

ASSESSMENT METHOD & ANALYSIS

Pre test was conducted on 27 March 2026.

Cell has been isolated at the basin level for collection of cold water directly from the rain zone.

All required data was collected and reports have been prepared for each test stage.

Since different conditions apply in each test, both pre and post tests have been compared to the design conditions. The difference between pre and post tests has been established accordingly.

INSTRUMENT PLACEMENT AND READINGS

Air flow was measured using Data Logging anemometer as per CTI ATC-143 Method of equal area — 10 traverses per quadrant (40 readings total for one fan).

Hot water temperature was taken at inlet of the hot water to the cooling tower.

Cold water temperature: 24 RTD sensors placed at air-inlet sides of the isolated cell (12 per side).

Water flow was measured using UFM (GE Make) on the riser.

WBT/DBT was measured using wet-bulb automatic stations recording every minute.

Power to the fan motor was noted from the client MCC.

CONCLUSION AND REMARKS

1. Testing of the cell was carried out after incorporating the distribution system changes on 27th of March 2026.
2. The additional head available caused flow to increase substantially. Even under these conditions the cell showed an improvement of approx. 0.5 °C above the post fan modification test.
3. Single cell isolated testing involves UNCERTAINTY in measurement due to turbulent water flow.

SUGGESTIONS AND SPECIAL NOTES

1. Improvement of 1.5 appears high in "TEST 2" so we consider that as 0.7 Deg C
2. Single cell isolated testing involves UNCERTAINTY in measurement due to turbulent water flow

FINAL DATA AND CONCLUSION OF PRE AND POST TEST

The following table presents a consolidated comparison of all measured and calculated parameters across the three test stages conducted during this evaluation.

Parameter	Unit	TEST 1 Pre-Test	TEST 2 Post Fan Change	TEST 3 Post Distr.
Water Flow	M3/hr	2998	3067.21	3680
WBT	Deg.C	25.25	24.22	21.7
HWT	Deg.C	44.67	43.21	42.13
CWT	Deg.C	35.08	32.89	32.4
Fan Power At Motor Inlet	KW	97.04	116.24	117
Range	Deg.C	9.59	10.32	9.73
Approach	Deg.C	9.83	8.67	10.7
CWT Deviation from Design	Deg.C	-14.82	-17.60	-22.11
Capability	%	395.1	373.2	359.0
Improvement vs Previous Test	Deg.C	—	2.77	4.51
Cumulative Improvement vs Test 1	Deg.C	—	2.77	7.28

NOTES

1. Improvement of 1.5 appears high in "TEST 2" so we consider that as 0.7 Deg C
2. Single cell isolated testing involves UNCERTAINTY in measurement due to turbulent water flow

ATC-105 PERFORMANCE EVALUATION — TEST 1 — PRE TEST

DESIGN VS RECORDED TEST CONDITIONS

Parameter	Unit	Design Conditions	Recorded Test Conditions
Water Flow	m³/hr	1000.00	2998.00
Hot Water Temp	°C	37.00	44.67
Cold Water Temp	°C	32.00	35.08
Wet Bulb Temp	°C	27.00	25.25
Range	°C	5.00	9.59
Approach	°C	5.00	9.83

STEP 1

TABLE 1 — CWT vs Range at Test WBT

Using the Merkel method, predicted Cold Water Temperatures (CWT) are computed at three fan speeds (90%, 100%, 110% of design flow) for three range values (80%, 100%, 120% of design range) at the recorded Test Wet Bulb Temperature.

Range %	Range (°C)	CWT @ 90% Flow	CWT @ 100% Flow	CWT @ 110% Flow
80%	4 °C	29.00	29.43	29.85
100%	5 °C	29.73	30.23	30.71
120%	6 °C	30.38	30.94	31.48

STEP 2

CROSS PLOT 1 — CWT vs Range at Test WBT

Plot the three CWT lines (90%, 100%, 110% flow) from Table 1 against Range. Draw a vertical line at the Test Range to read off the CWT for each flow condition. These three (Flow, CWT) pairs form Table 2.

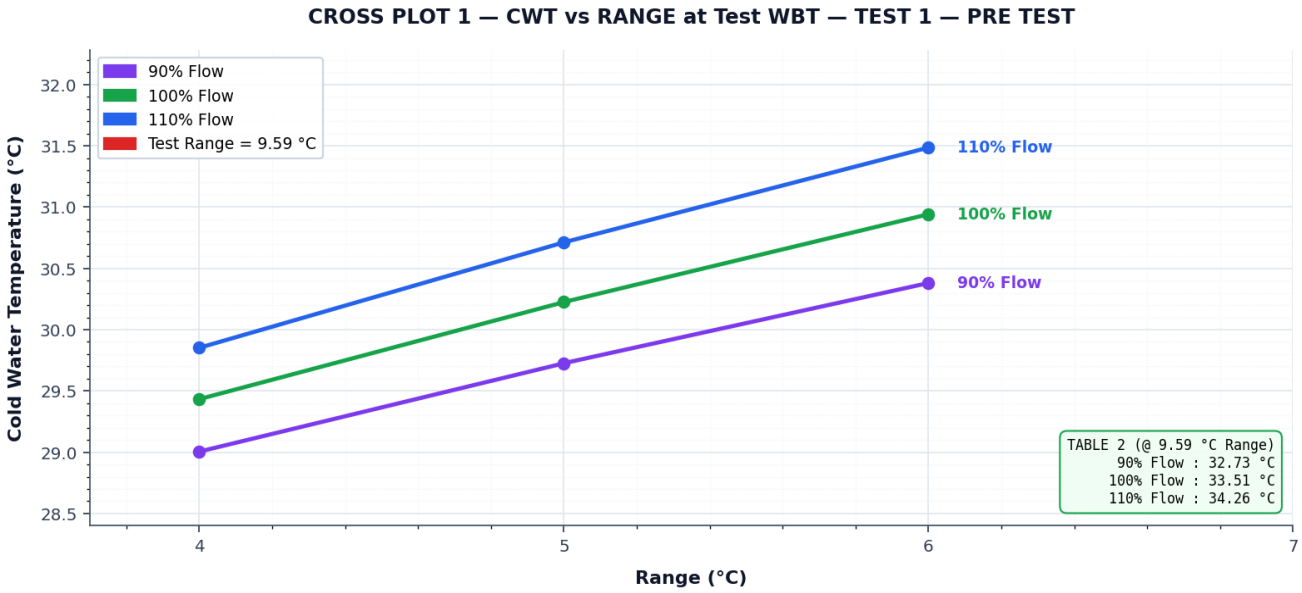

Figure 1: Cross Plot 1 — CWT vs Range at Test WBT — TEST 1 — PRE TEST

TABLE 2 — CWT at Test Range = 9.59 °C

Flow Condition	Water Flow (m³/hr)	CWT at Range 9.59°C
90% Flow	900.00	32.73
100% Flow	1000.00	33.51
110% Flow	1100.00	34.26

STEP 3 — ADJUSTED WATER FLOW — Air Density Correction

Adjust the recorded water flow for air density differences between test and design conditions: Q_adj = Q_test × (Air Density Ratio)^0.5. The density ratio accounts for atmospheric variation in air mass flow delivered by the fan.

i. Recorded Test Water Flow (Q_test)	**2998.00 m³/hr**
ii. Air Density Ratio (ρ_design / ρ_test)	**1.00**
iii. CWT @ 90% Design Flow (Cross Plot 1)	**32.73 °C**
iv. CWT @ 100% Design Flow (Cross Plot 1)	**33.51 °C**
v. CWT @ 110% Design Flow (Cross Plot 1)	**34.26 °C**
vi. Test Hot Water Temperature (HWT)	**44.67 °C**
vii. Test Cold Water Temperature (CWT)	**35.08 °C**
viii. Test Wet Bulb Temperature (WBT)	**25.25 °C**
ix. Fan Power — Design	**117.00 kW**
x. Fan Power — Test	**97.04 kW**
Q_adj — Adjusted Water Flow (Density Corrected)	**3188.80 m³/hr**

STEP 4 — CROSS PLOT 2 — Water Flow vs CWT (Design WBT & Range)

Plot (Water Flow, CWT) pairs from Table 2. Locate Q_adj on the performance curve to read the Predicted CWT. Then find the water flow at which the curve intersects the Design CWT — this is the Predicted Flow used in Step 5.

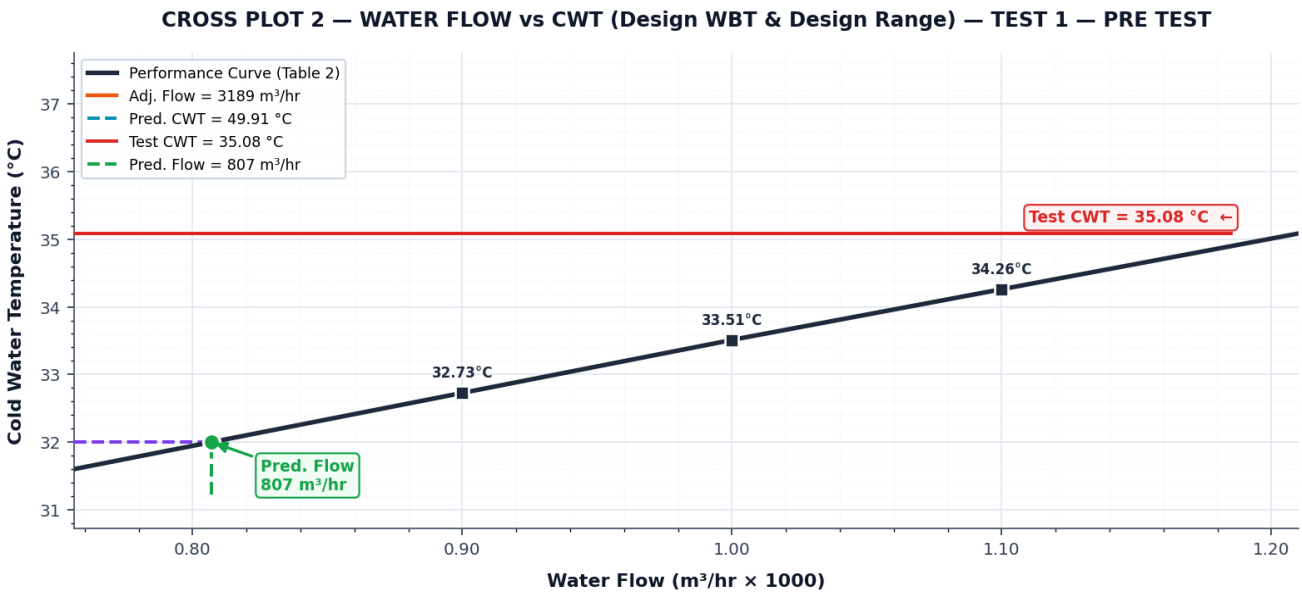

Figure 2: Cross Plot 2 — Water Flow vs CWT (Design WBT & Range) — TEST 1 — PRE TEST

FINAL ATC-105 RESULTS — TEST 1 — PRE TEST

STEP 5

FINAL EVALUATION — CWT Shortfall & Capability

Compare the Predicted CWT (from Cross Plot 2 at Q_adj) with the Test CWT. CWT Shortfall = Test CWT – Predicted CWT. A negative shortfall means the tower performed below specification. Capability % = (Q_adj / Predicted Flow) × 100.

Step	Parameter	Value
i	Q_adj — Adjusted Water Flow (density corrected)	3188.80 m³/hr
ii	Predicted CWT — from Cross Plot 2 at Q_adj	49.91 °C
iii	Test CWT — Recorded during test	35.08 °C
iv	Predicted Water Flow — at Design CWT on Cross Plot 2	807.02 m³/hr
v	CWT Shortfall = Test CWT – Predicted CWT	-14.82 °C
vi	Cooling Tower Capability = (Q_adj / Pred. Flow) × 100	395.10 %
vii	Air Density Ratio applied	1.00

CONCLUSION — TEST 1 — PRE TEST

CWT SHORTFALL IN PERFORMANCE

-14.82 °C

Test CWT (35.08°C) vs Predicted CWT (49.91°C)

COOLING TOWER CAPABILITY

395.1 %

= 3189 / 807 × 100

Tower is BELOW specification by 14.82°C — Capability: 395.1%

ATC-105 PERFORMANCE EVALUATION — TEST 2 — POST FAN CHANGE

DESIGN VS RECORDED TEST CONDITIONS

Parameter	Unit	Design Conditions	Recorded Test Conditions
Water Flow	m³/hr	1000.00	3067.21
Hot Water Temp	°C	37.00	43.21
Cold Water Temp	°C	32.00	32.89
Wet Bulb Temp	°C	27.00	24.22
Range	°C	5.00	10.32
Approach	°C	5.00	8.67

STEP 1

TABLE 1 — CWT vs Range at Test WBT

Using the Merkel method, predicted Cold Water Temperatures (CWT) are computed at three fan speeds (90%, 100%, 110% of design flow) for three range values (80%, 100%, 120% of design range) at the recorded Test Wet Bulb Temperature.

Range %	Range (°C)	CWT @ 90% Flow	CWT @ 100% Flow	CWT @ 110% Flow
80%	4 °C	28.21	28.66	29.10
100%	5 °C	28.98	29.50	30.00
120%	6 °C	29.67	30.25	30.82

STEP 2

CROSS PLOT 1 — CWT vs Range at Test WBT

Plot the three CWT lines (90%, 100%, 110% flow) from Table 1 against Range. Draw a vertical line at the Test Range to read off the CWT for each flow condition. These three (Flow, CWT) pairs form Table 2.

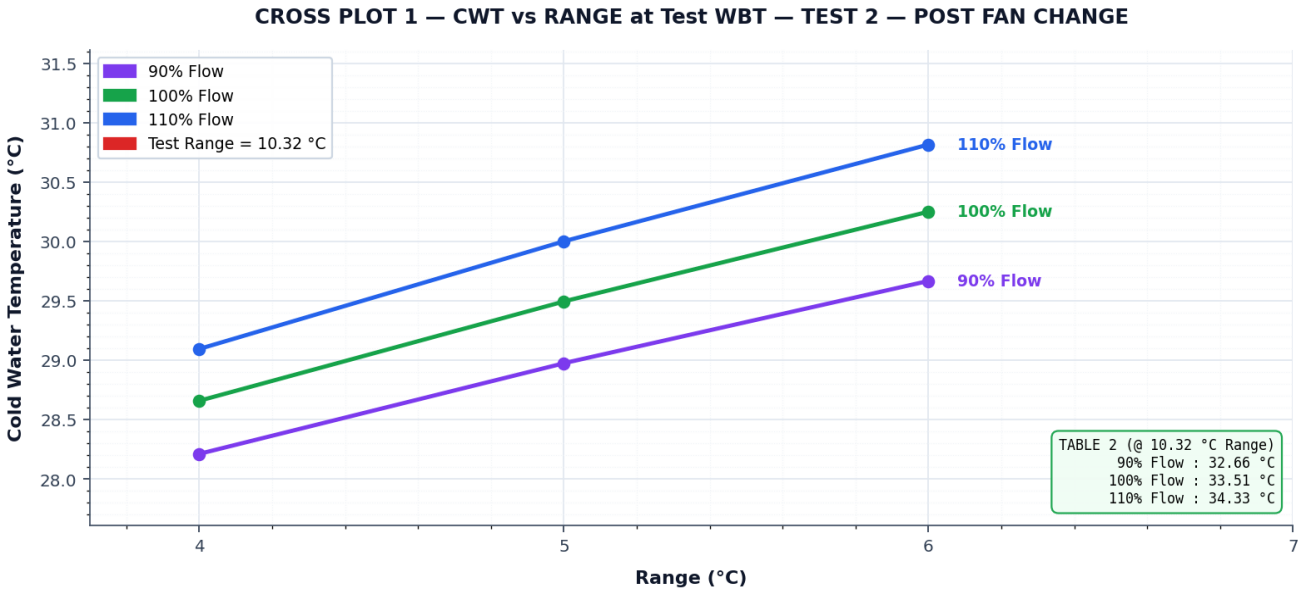

Figure 1: Cross Plot 1 — CWT vs Range at Test WBT — TEST 2 — POST FAN CHANGE

TABLE 2 — CWT at Test Range = 10.32 °C

Flow Condition	Water Flow (m³/hr)	CWT at Range 10.32°C
90% Flow	900.00	32.66
100% Flow	1000.00	33.51
110% Flow	1100.00	34.33

STEP 3 — ADJUSTED WATER FLOW — Air Density Correction

Adjust the recorded water flow for air density differences between test and design conditions: Q_adj = Q_test × (Air Density Ratio)^0.5. The density ratio accounts for atmospheric variation in air mass flow delivered by the fan.

i. Recorded Test Water Flow (Q_test)	**3067.21 m³/hr**
ii. Air Density Ratio (ρ_design / ρ_test)	**1.00**
iii. CWT @ 90% Design Flow (Cross Plot 1)	**32.66 °C**
iv. CWT @ 100% Design Flow (Cross Plot 1)	**33.51 °C**
v. CWT @ 110% Design Flow (Cross Plot 1)	**34.33 °C**
vi. Test Hot Water Temperature (HWT)	**43.21 °C**
vii. Test Cold Water Temperature (CWT)	**32.89 °C**
viii. Test Wet Bulb Temperature (WBT)	**24.22 °C**
ix. Fan Power — Design	**117.00 kW**
x. Fan Power — Test	**116.24 kW**
Q_adj — Adjusted Water Flow (Density Corrected)	**3072.59 m³/hr**

STEP 4 — CROSS PLOT 2 — Water Flow vs CWT (Design WBT & Range)

Plot (Water Flow, CWT) pairs from Table 2. Locate Q_adj on the performance curve to read the Predicted CWT. Then find the water flow at which the curve intersects the Design CWT — this is the Predicted Flow used in Step 5.

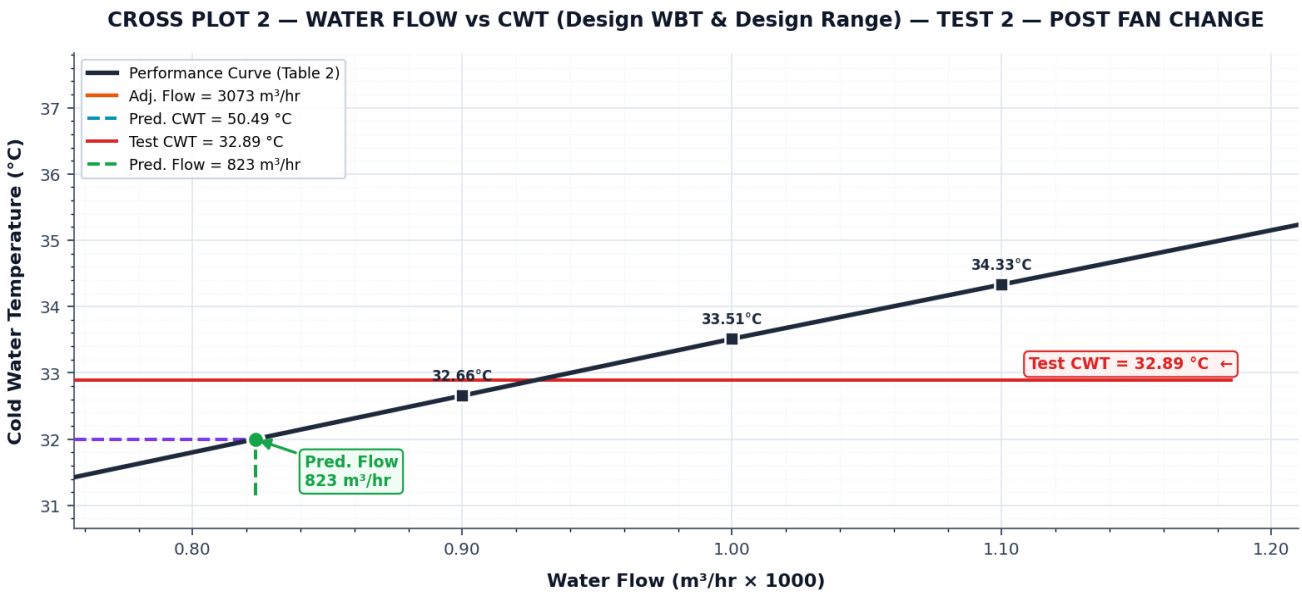

Figure 2: Cross Plot 2 — Water Flow vs CWT (Design WBT & Range) — TEST 2 — POST FAN CHANGE

FINAL ATC-105 RESULTS — TEST 2 — POST FAN CHANGE

STEP 5

FINAL EVALUATION — CWT Shortfall & Capability

Compare the Predicted CWT (from Cross Plot 2 at Q_adj) with the Test CWT. CWT Shortfall = Test CWT – Predicted CWT. A negative shortfall means the tower performed below specification. Capability % = (Q_adj / Predicted Flow) × 100.

Step	Parameter	Value
i	Q_adj — Adjusted Water Flow (density corrected)	3072.59 m³/hr
ii	Predicted CWT — from Cross Plot 2 at Q_adj	50.49 °C
iii	Test CWT — Recorded during test	32.89 °C
iv	Predicted Water Flow — at Design CWT on Cross Plot 2	823.25 m³/hr
v	CWT Shortfall = Test CWT – Predicted CWT	-17.60 °C
vi	Cooling Tower Capability = (Q_adj / Pred. Flow) × 100	373.20 %
vii	Air Density Ratio applied	1.00

CONCLUSION — TEST 2 — POST FAN CHANGE

CWT SHORTFALL IN PERFORMANCE	COOLING TOWER CAPABILITY
-17.60 °C	**373.2 %**
Test CWT (32.89°C) vs Predicted CWT (50.49°C)	= 3073 / 823 × 100

Tower is BELOW specification by 17.60°C — Capability: 373.2%

ATC-105 PERFORMANCE EVALUATION — TEST 3 — POST DISTRIBUTION CHANGE

DESIGN VS RECORDED TEST CONDITIONS

Parameter	Unit	Design Conditions	Recorded Test Conditions
Water Flow	m³/hr	1000.00	3680.00
Hot Water Temp	°C	37.00	42.13
Cold Water Temp	°C	32.00	32.40
Wet Bulb Temp	°C	27.00	21.70
Range	°C	5.00	9.73
Approach	°C	5.00	10.70

STEP 1

TABLE 1 — CWT vs Range at Test WBT

Using the Merkel method, predicted Cold Water Temperatures (CWT) are computed at three fan speeds (90%, 100%, 110% of design flow) for three range values (80%, 100%, 120% of design range) at the recorded Test Wet Bulb Temperature.

Range %	Range (°C)	CWT @ 90% Flow	CWT @ 100% Flow	CWT @ 110% Flow
80%	4 °C	26.31	26.80	27.28
100%	5 °C	27.18	27.75	28.30
120%	6 °C	27.97	28.61	29.22

STEP 2

CROSS PLOT 1 — CWT vs Range at Test WBT

Plot the three CWT lines (90%, 100%, 110% flow) from Table 1 against Range. Draw a vertical line at the Test Range to read off the CWT for each flow condition. These three (Flow, CWT) pairs form Table 2.

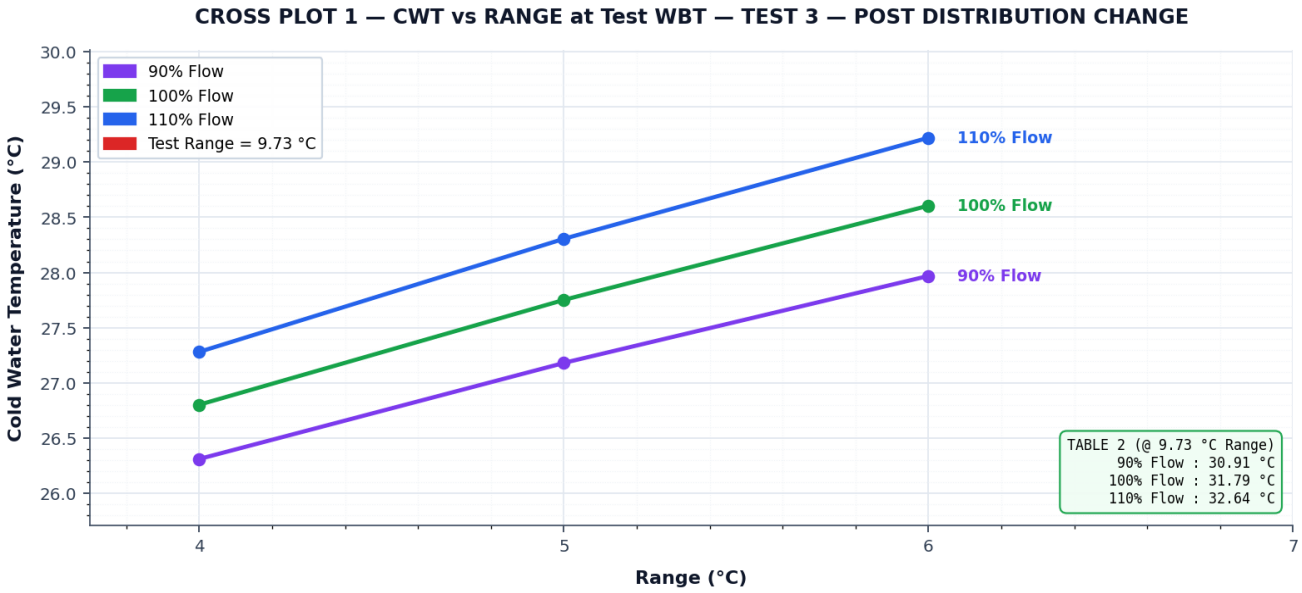

Figure 1: Cross Plot 1 — CWT vs Range at Test WBT — TEST 3 — POST DISTRIBUTION CHANGE

TABLE 2 — CWT at Test Range = 9.73 °C

Flow Condition	Water Flow (m³/hr)	CWT at Range 9.73°C
90% Flow	900.00	30.91
100% Flow	1000.00	31.79
110% Flow	1100.00	32.64

STEP 3 — ADJUSTED WATER FLOW — Air Density Correction

Adjust the recorded water flow for air density differences between test and design conditions: Q_adj = Q_test × (Air Density Ratio)^0.5. The density ratio accounts for atmospheric variation in air mass flow delivered by the fan.

i. Recorded Test Water Flow (Q_test)	**3680.00 m³/hr**
ii. Air Density Ratio (ρ_design / ρ_test)	**1.00**
iii. CWT @ 90% Design Flow (Cross Plot 1)	**30.91 °C**
iv. CWT @ 100% Design Flow (Cross Plot 1)	**31.79 °C**
v. CWT @ 110% Design Flow (Cross Plot 1)	**32.64 °C**
vi. Test Hot Water Temperature (HWT)	**42.13 °C**
vii. Test Cold Water Temperature (CWT)	**32.40 °C**
viii. Test Wet Bulb Temperature (WBT)	**21.70 °C**
ix. Fan Power — Design	**117.00 kW**
x. Fan Power — Test	**117.00 kW**
Q_adj — Adjusted Water Flow (Density Corrected)	**3678.80 m³/hr**

STEP 4 — CROSS PLOT 2 — Water Flow vs CWT (Design WBT & Range)

Plot (Water Flow, CWT) pairs from Table 2. Locate Q_adj on the performance curve to read the Predicted CWT. Then find the water flow at which the curve intersects the Design CWT — this is the Predicted Flow used in Step 5.

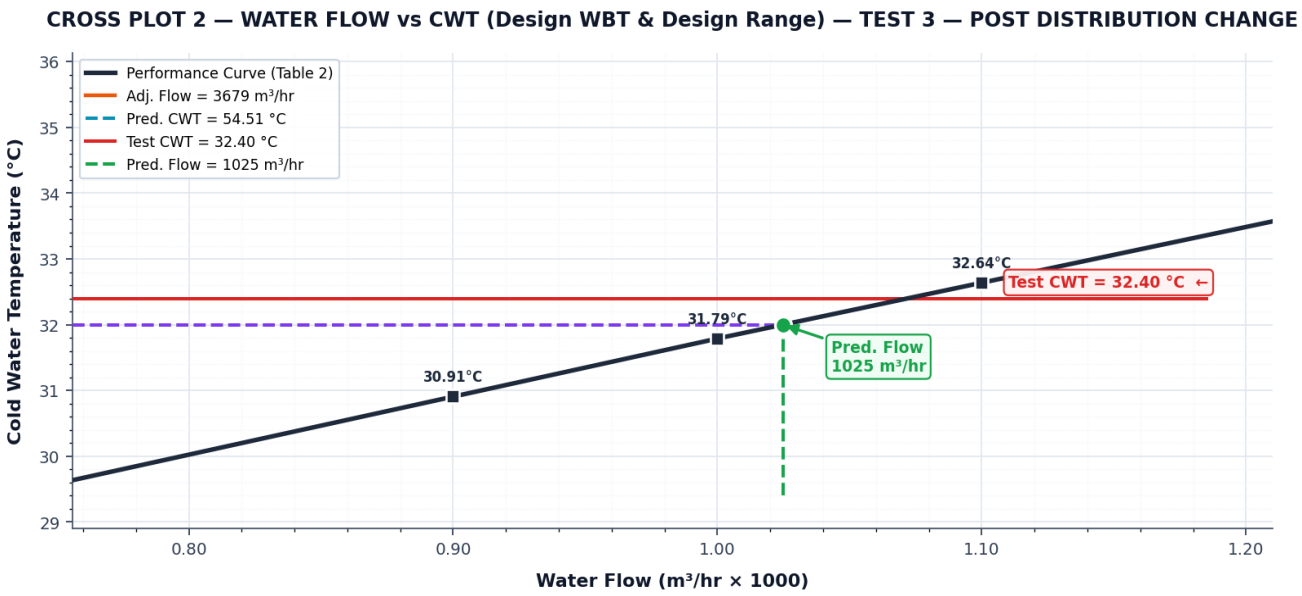

Figure 2: Cross Plot 2 — Water Flow vs CWT (Design WBT & Range) — TEST 3 — POST DISTRIBUTION CHANGE

FINAL ATC-105 RESULTS — TEST 3 — POST DISTRIBUTION CHANGE

STEP 5

FINAL EVALUATION — CWT Shortfall & Capability

Compare the Predicted CWT (from Cross Plot 2 at Q_adj) with the Test CWT. CWT Shortfall = Test CWT – Predicted CWT. A negative shortfall means the tower performed below specification. Capability % = (Q_adj / Predicted Flow) × 100.

Step	Parameter	Value
i	Q_adj — Adjusted Water Flow (density corrected)	3678.80 m³/hr
ii	Predicted CWT — from Cross Plot 2 at Q_adj	54.51 °C
iii	Test CWT — Recorded during test	32.40 °C
iv	Predicted Water Flow — at Design CWT on Cross Plot 2	1024.76 m³/hr
v	CWT Shortfall = Test CWT – Predicted CWT	-22.11 °C
vi	Cooling Tower Capability = (Q_adj / Pred. Flow) × 100	359.00 %
vii	Air Density Ratio applied	1.00

CONCLUSION — TEST 3 — POST DISTRIBUTION CHANGE

CWT SHORTFALL IN PERFORMANCE

-22.11 °C

Test CWT (32.40°C) vs Predicted CWT (54.51°C)

COOLING TOWER CAPABILITY

359.0 %

= 3679 / 1025 × 100

Tower is BELOW specification by 22.11°C — Capability: 359.0%

——— END OF REPORT ———

SS COOLING TOWER CONSULTANTS
www.ssctc.org
15 April 2026